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Paleomagnetic study of some of the dikes from the Esh El Mellaha Range, Eastern Desert, Egypt

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Abstract—Two oppositely directed secondary directions of magnetization were isolated from some of the dikes in the Esh El Mellaha Range. Radiometric and paleomagnetic age considerations suggest that these were acquired during an interval of low-grade metamorphism in late Cambrian time (approx. 530 m.y.). The paleopole position for the acidic dikes (three sites) is 87.3N 25.9E dp 9.6, dm 14.5, for the intermediate dikes (eight sites) 81.6N 120.8E dp 3.3 dm 5.2 and for the basic dikes (11 sites) 82.7N 86.5W dp 5.6 dm 9.1.

INTRODUCTION

THE ESH EL MELLAHA Range forms a long, low and narrow range of hills which run parallel to the Red Sea margin. They extend for about 75 km from south of Ras Gharib to just north of Hurghada (Fig. 1) and at their broadest point are just over 6 km wide. The range rises above the coastal plain to the east, and to the west is separated by a wide valley from the main mass of the Red Sea Hills. Both areas are covered by recent sediments. The Esh El Mellaha Range is probably a tilted horst block similar to Gebel Zeit but one which has been more deeply buried by recent sediments. This is conjectural, however, for the boundary faults are mapped along only part of the length of the range (Fig. 2). The principal geological works on the Esh El Mellaha Range are by Kabesh and Shahin (1968), Kabesh *et al.* (1970a,b) and Lotfi Abdel-Khalek and Hafez (1973).

Basement rocks crop out mainly in the eastern part of the range where there are isolated Miocene outliers. The western part is mantled by Cretaceous and younger sediments (Fig. 2). Good exposures of the rocks of basement complex are found in many of the wadis which cross the range from southwest to northeast, in particular the wadis El Mellaha, Dib, Abu Had and Betih.

The basement rocks are cut by many dikes. Kabesh and Shahin (1968) recorded 500, but that is a minimum. Although the authors recognized about 62 petrological types, they were able to group them as acid, intermediate, alkaline and basic in a generalized oldest to youngest age sequence based upon cross-cutting relationships.

A suite of the dikes was collected during the 1980 field season as part of a project of studying the minor intrusions of the Eastern Desert with interest centred around the possibility of using the relative age sequence to provide pole positions for successive time intervals which could be compared with dike data presented by Nairn *et al.* (1980), Ressetar and Nairn (1980), and with work on the Dokhan volcanic series (Perry 1982) and the Aswan dikes of Yomede (1981).

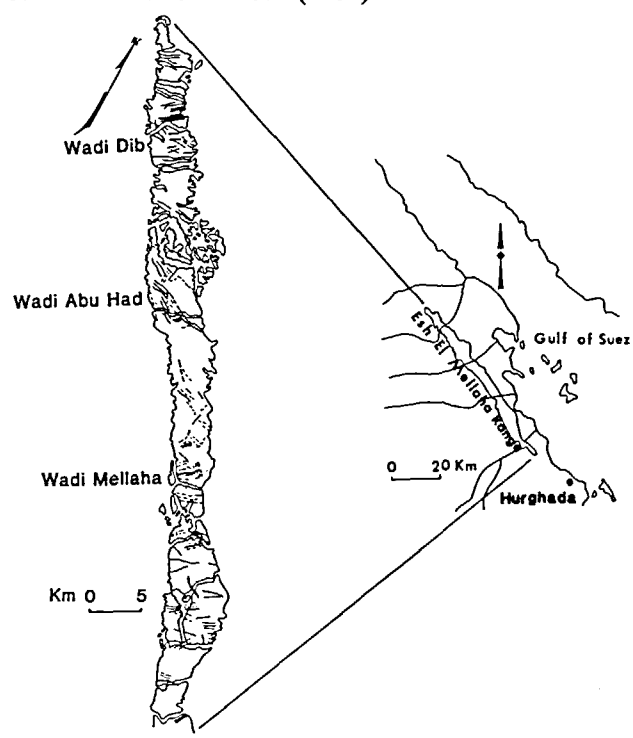


Fig. 1. Location map of the Esh El Mellaha Range, Eastern Desert, showing some of the major dikes to illustrate the dike trends.

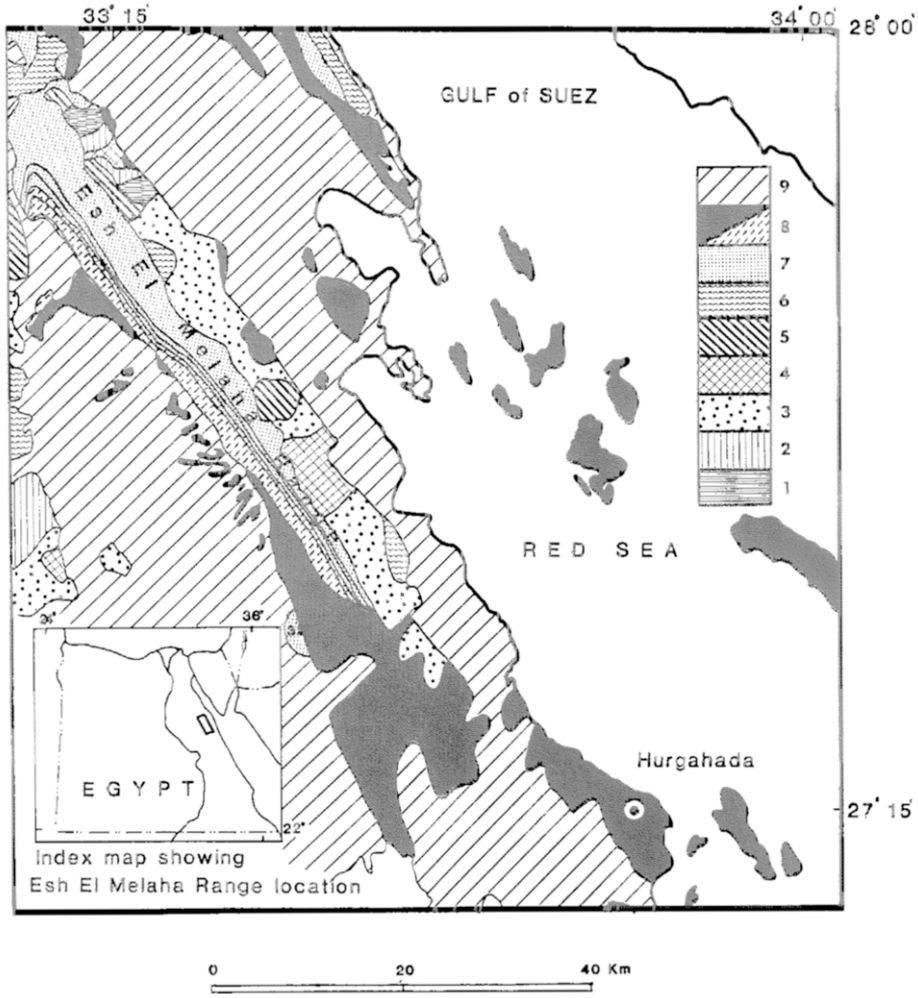


Fig. 2. Simplified geological map of the Esh El Mellaha Range. (1) Pre-Cambrian geosynclinal metasediments, (2) Pre-Cambrian Shadly volcanics, (3) Pre-Cambrian older granites, (4) "Pre-Cambrian" Dokhan volcanics, (5) Hammamat Group, (6) Gattarian younger granitoids, (7) Upper Cretaceous (undifferentiated), (8) Tertiary (a) Pliocene–M. Miocene marine to paralic deposits (b) L. Eocene limestone and Paleocene shale and marl, (9) Quaternary alluvial and beach deposits.

Even in the Red Sea area, which Vail (1970) considered to be one of the most heavily intruded regions in the world, the number of dikes in the Esh El Mellaha Range is exceptional. The sketch map (Fig. 1) shows sufficient numbers only to help delineate trends, since it is impossible to show them all. As Kabesh and Shahin (1968) point out, the dikes are most clearly seen where basic dikes cut through granites or where acidic dikes penetrate dark-coloured rocks such as the Dokhan volcanics. Basic dikes cutting the latter formation are harder to distinguish and tend, therefore, to be under-represented. The principal dike trends are NE–SW or E–W. Two other trends which exist but are less commonly found are N–S and NW–SE.

The dikes are generally vertical or steeply inclined (80–85°) and tend to occur in parallel swarms. They are generally simple, Kabesh and Shahin (1968) report very few composite dikes. They may vary in thickness from a few cm to several km. The average width found during our sampling program, measured on about 70 dikes, was 2 m. An order of magnitude calculation would suggest that the dike intrusions represent a minimum extension of the range of 3–5%.

The age of the dikes is only known within broad limits. They are certainly pre-Nubian and equally certainly represent the last igneous phase to affect basement rocks (Kabesh *et al.* 1970a,b). Although not dated in the Esh El Mellaha Range, the granites found there are equated with the Younger Granites, for which ages in the 500–600 m.a. range have been proposed by Hashad *et al.* (1972) and Fullagar and Greenberg (1978). This would imply a late Cambrian age or younger for the dikes. A complication is introduced by the uncertainty of the age of the Hammamat formation for, on the basis of faunal descriptions on collections made in the Wadi Hammamat, Soliman (1980) considered part of the Hammamat to be as young as Devonian.

At our request Dr. R. E. Dooley carried out K/Ar analyses and provided ages from a single sample from each of the major dike suites as shown in Table 1.

Table 1

		K%	Ar	App. age
Basic dike site 44		1.41	1.4863	524
	duplicate	1.391	1.3935	532
Acidic dike site 42		1.282	1.2605	500
Intermediate dike site 37		6.054	5.0352	425
	duplicate	6.155	5.1954	431

The ages are minimum ages, and it seems probable that the most acceptable are those from the basic and acidic units, which appear to be petrologically the least altered. Both petrographic and paleomagnetic studies suggest that the intermediate (andesite) rocks have been most altered and hence argon loss makes the low ages suspect. As might be expected there is a rough correlation between the age and the degree of alteration so that the apparent contradiction between these age relationships and that determined on field evidence is not significant.

PETROGRAPHIC OUTLINE

Thin sections of some of the dikes were examined as a control on the field classification into basic, acid and intermediate dikes. Attention was given to the degree of alteration, for all dikes were found to be considerably altered, and to the state of the ferromagnetic minerals.

In hand sample or cores, the basic dikes tend to be dark or blackish in colour. In thin section they tend to be slightly porphyritic with pyroxene, amphibole or felspar phenocrysts. The ferromagnesian minerals are usually chloritized, sometimes to the extent that the original minerals cannot be identified. The felspar both in phenocryst and groundmass is largely sericitized. The abundant opaque minerals appear to be primary; corrosion rims are absent; leucoxene, though present, is not abundant. Fine-grained secondary hematite is seen penetrating along cracks.

The intermediate rocks are bluish in hand sample. In thin section they appear to be predominantly andesitic. As in the case of the basic rocks, the felspar and ferromagnesian minerals are largely chloritized and sericitized, and the association of this with chlorite, sphene and epidote suggests the dikes have been subjected to low-grade metamorphism and altered hydrothermally. Hematite is more abundant than in the basic rocks, being found both as discrete crystals as well as filling fissures.

According to Kabesh *et al.* (1970b) the acidic dikes may be grouped as granitic, granophyric and felsic. The sites we examined appear to be porphyritic, fine-grained, hypidiomorphic granites. The felspar is sericitized and there is evidence of secondary hematite. The alteration, although extensive, seems less extensive than in the other two groups of dikes.

PALEOMAGNETIC STUDY

An average of six samples per dike was collected from natural outcrops along the walls of wadis through the Esh El Mellaha Range. Although sampling was more concentrated in the central and southern parts of the range, all parts were visited and sampled. Most were collected using a gasoline-powered field drill, but some were block samples later cored in the laboratory. Samples were spaced at about 1 m intervals and distributed across the dike. They were oriented with a device incorporating a Brunton compass. No deviations were noted when compared with orienting with the compass more distant from the dike. They were sliced to a standard length in the laboratory, most commonly giving one core per sample.

The natural remanent magnetization (NRM) of the cores was measured using a "Digico complete results magnetometer" interfaced with a Digital MINC computer. There was a considerable range in the intensity of NRM ranging from values of 8×10^{4} mAm^{-1} to about 1.3×10^{8} mAm^{-1} in the basic dikes, to a maximum of 1.5×10^{-4} mAm^{-1} in the intermediate dikes, down to 1×10^{-3} to 3×10^{-1} mAm^{-1} in the acidic rocks. Progressive alternating field demagnetization of the natural remanent magnetization in a Schonstedt GSD-1 was carried out in 5 mT steps to 25 mT, above that at 35, 50, 75 and 100 mT on two pilot cores per site (Fig. 3).

Thermal demagnetization using the Schonstedt TD was carried out on some samples. The first stages of heating were generally at widely spaced intervals beginning at 200°C and then at 100°C steps, but as higher temperatures were reached the interval was reduced to 50°C (Fig. 4).

Programs available on the MINC allowed us to calculate the usual statistical parameters and to compute the vector of the magnetization removed between progressive demagnetization steps. This proved necessary in some samples which have superposed magnetic vectors.

DISCUSSION OF RESULTS

(1) *Directions of remanence*

The measurements of the direction of remanence during progressive demagnetization permit us to distinguish three patterns of behaviour. For most of the acid and basic dikes examined, and about half of the andesites, after the elimination of a small viscous component there appeared to be but a single component, for the direction of magnetization remained virtually unchanged during demagnetization. This behaviour is illustrated through stable end point curves (Figs 5 and 6) and is the same whether alternating field or thermal techniques were applied. These curves also illustrate the point that in some samples the directions of magnetization do not always decay exactly toward the origin, although the directions of magnetization are identical to those of other samples from the same site which do. In some cases an improvement occurred when thermal treatment was applied to samples demagnetized in alternating fields (BM 372, Fig. 7).

At the other end of the spectrum there were samples, sometimes all samples from a given site, where no stable end point was reached, and these had to be excluded. This behaviour was found in samples from all lithologies and could not be predicted on any obvious lithological basis.

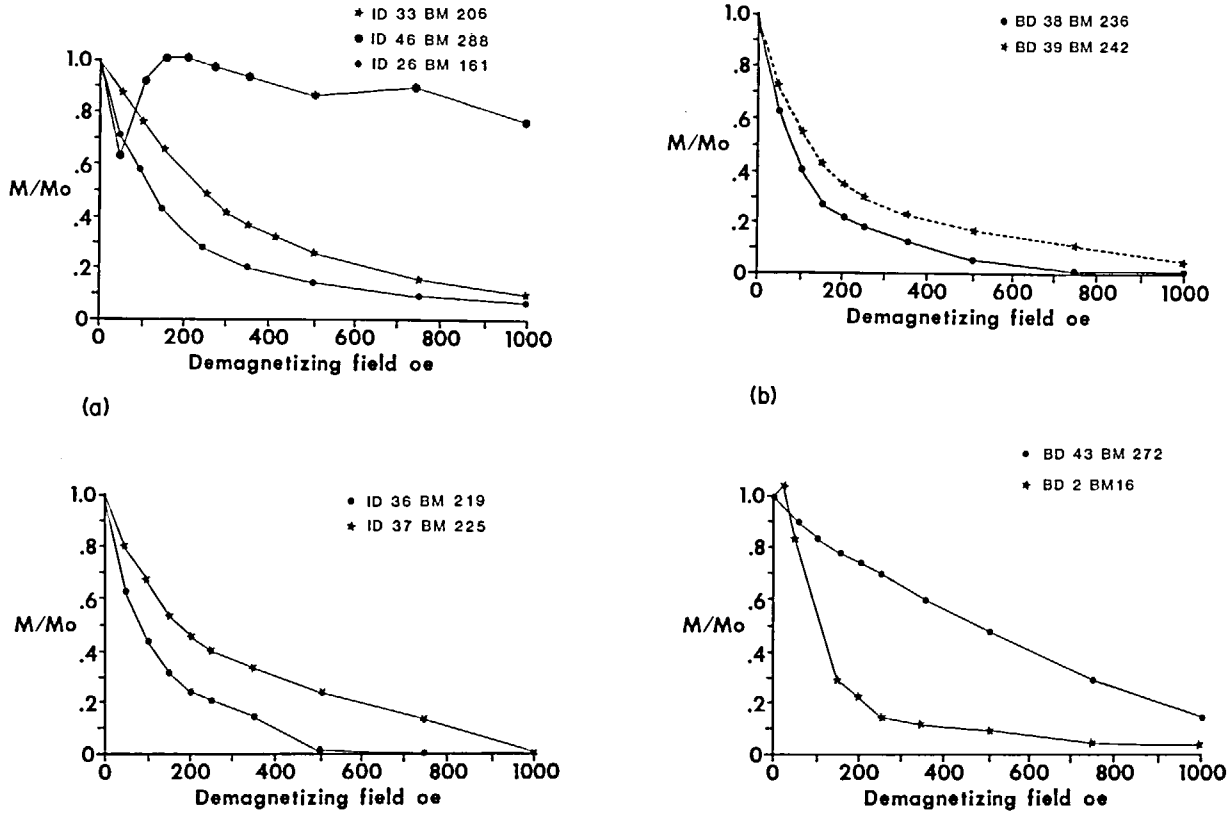


Fig. 3. (a) and (b) Normalized demagnetization M/M_0 curves from samples treated in alternating fields (oe = oersted or 10^{-1}mT).

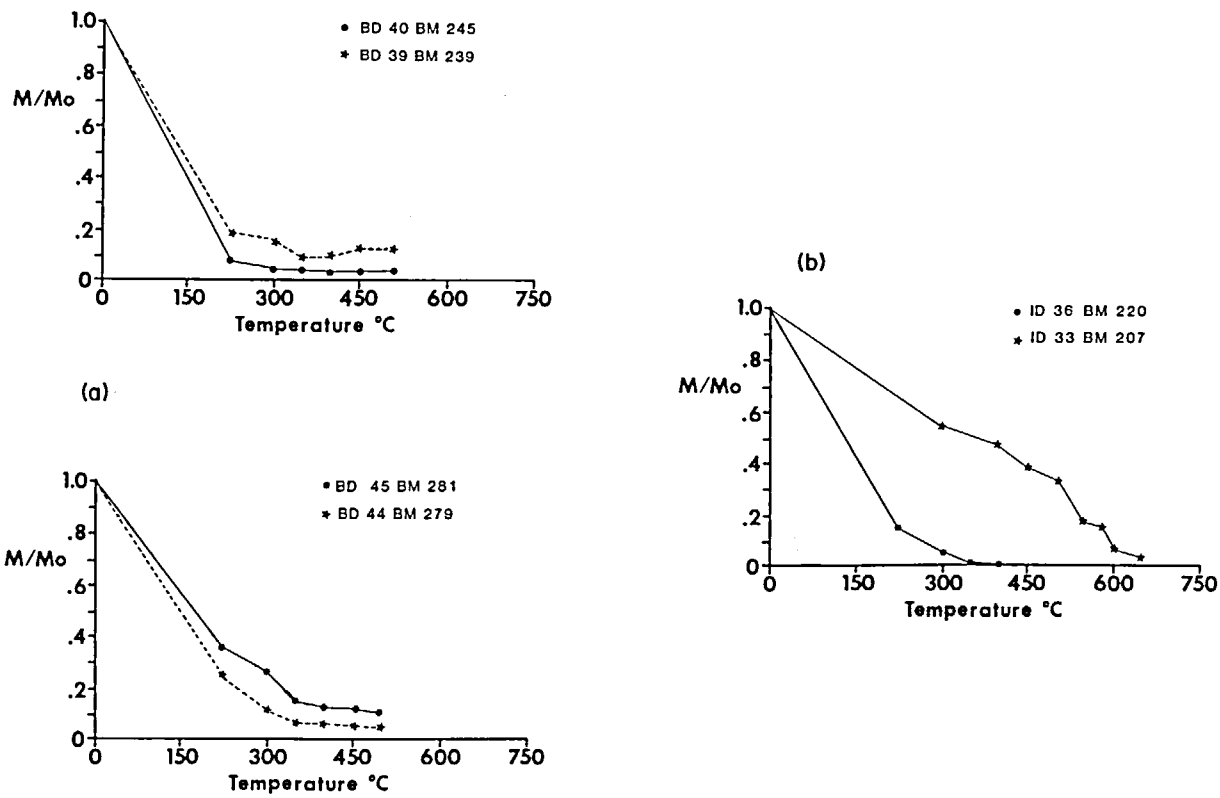


Fig. 4. (a) and (b) Normalized demagnetization curves from thermally demagnetized samples.

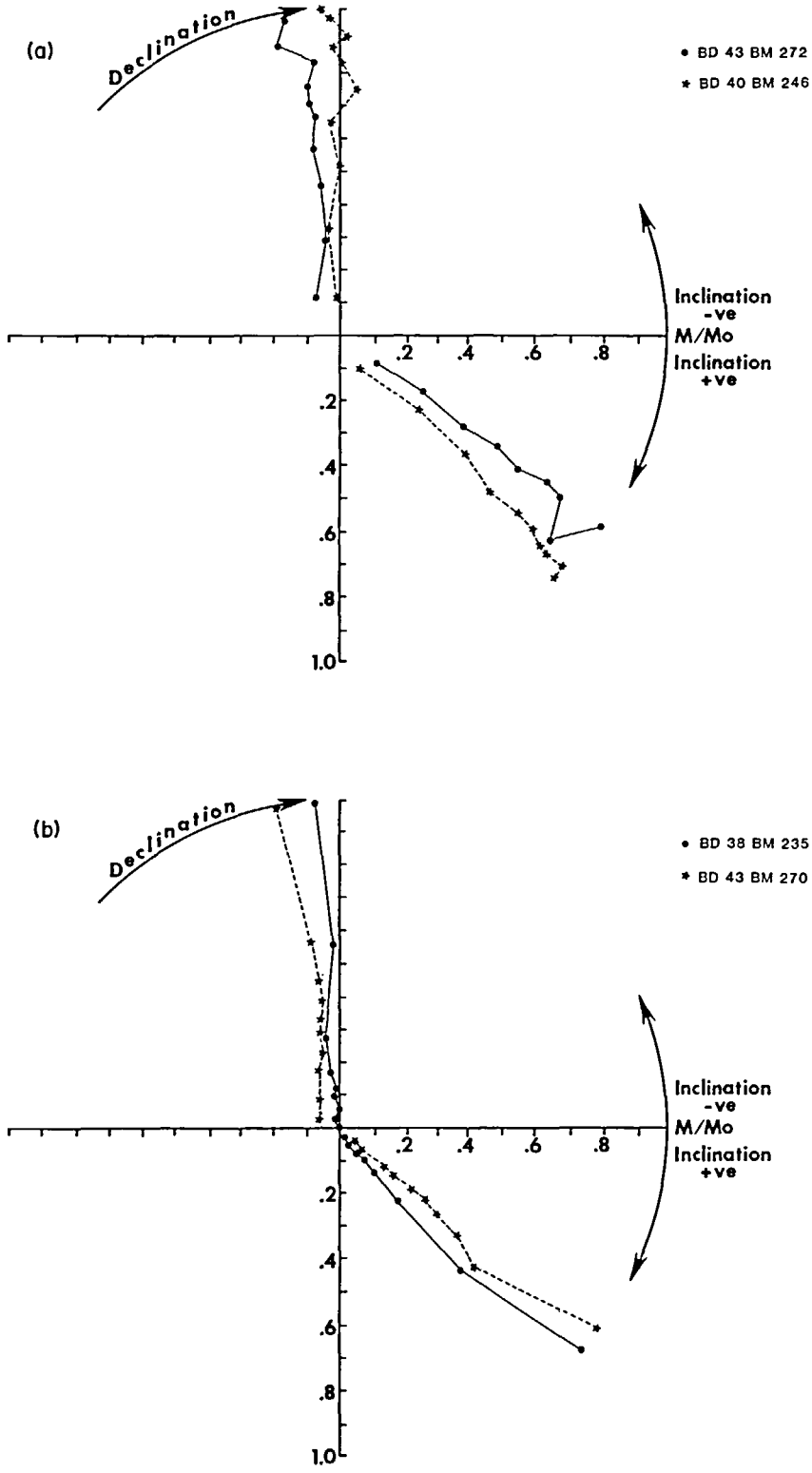


Fig. 5. (a) and (b) Stable end point curves in which normalized intensity is plotted against declination and inclination.

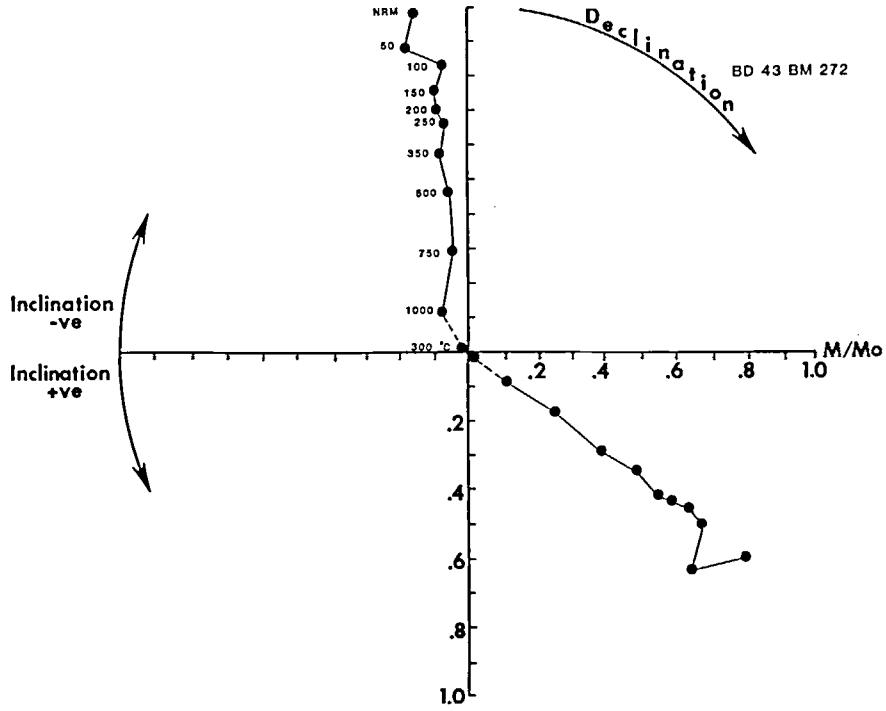


Fig. 6. Stable end point curve plotted as in Fig. 5 showing the effect of one stage of thermal demagnetization following alternating field treatment.

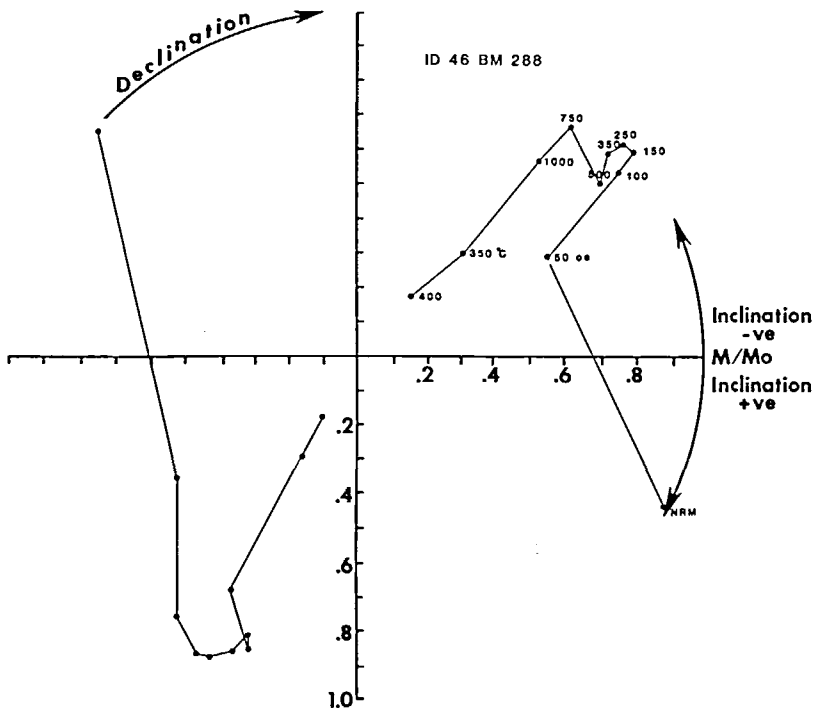


Fig. 7. Stable end point curve of sample BM 288 showing the effects of combined alternating field and thermal demagnetization.

A third group of samples, consisting mainly of andesites, showed clear evidence of superposed magnetizations associated with magnetic components with overlapping coercivity spectra. The clearest example of this behaviour is seen in sample BM 288 (from site 1D46). As shown in Fig. 3a, after treatment in fields of 100 mT about 80% of the NRM intensity remained and the directions of magnetization showed little tendency to migrate toward the origin (Fig. 8) until thermally demagnetized. Thermal demagnetization permitted us to isolate a reversed direction of magnetization for that sample and site. The directions of magnetization of the vector removed proved to be normally magnetized and the same as that recovered from dikes showing only a single component. In other samples the vector removed was essentially the same, and at successive stages the remaining vector migrated toward the reversed direction (Fig. 8). We were able to identify the reversed magnetization in 12 samples from three sites, and calculated a mean direction of magnetization (Table 2d). In other samples, no clear pattern of behaviour emerged.

The site mean directions of magnetism are tabulated (Table 2) and figured (stereograms Fig. 9).

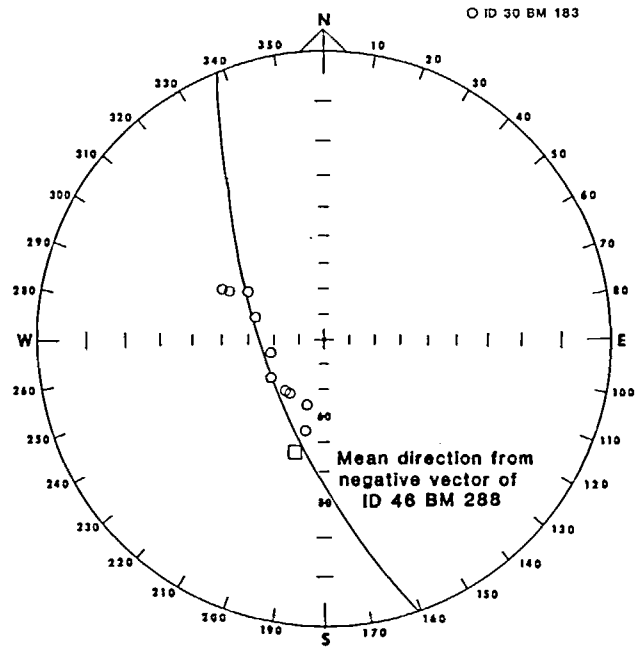


Fig. 8. Stereographic plot showing the migration of the negative magnetization vector during progressive demagnetization.

(2) *The age of the magnetization*

Although the K/Ar age dates on the dikes are few, and the granites of the Esh El Mellaha Range have not been dated, a reasonable case can be argued geologically for an age in the range 480–580 m.a. The ages for the acid and basic dikes are regarded as minimum ages, and because of the generally high degree of alteration of the andesites, argon loss may be suspected. We associate that alteration with the acquisition of the secondary, hard, magnetic component. The question that remains is whether the normal magnetization is a thermo-remanent magnetization or if it, too, is secondary. Since the reversed direction is antipodal to the normal magnetization, and since the degree of alteration in all the dikes is high, we conclude that normal magnetization is also secondary. Thus the ages and the magnetization appear to have been acquired at the time of alteration of the

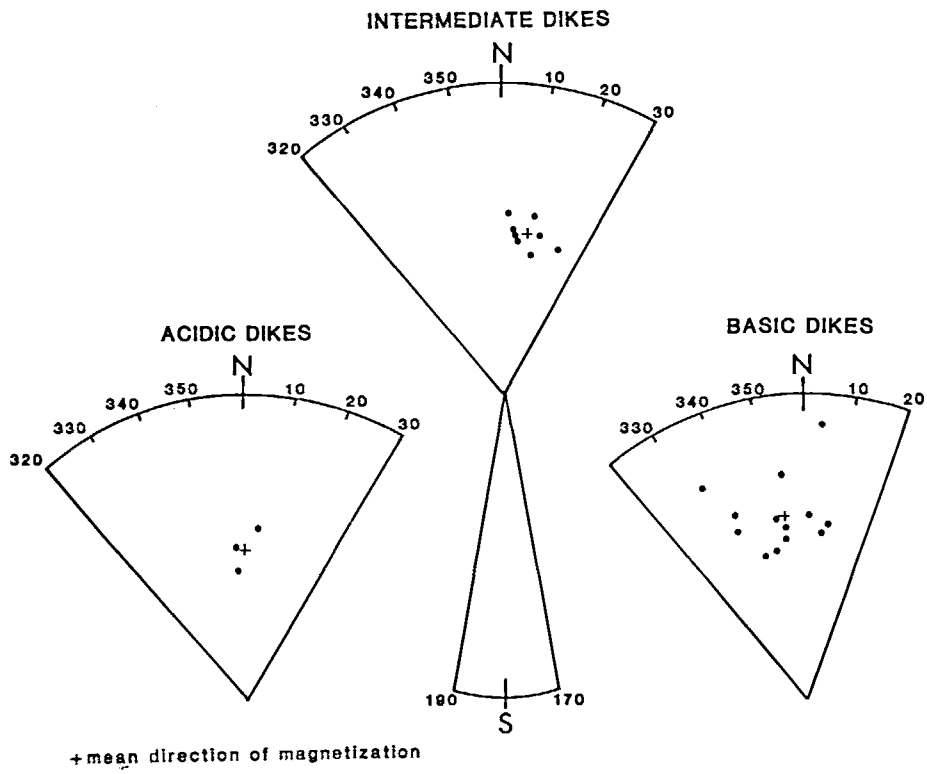


Fig. 9. Site mean directions of magnetization after magnetic cleaning shown by group.

Table 2. Summary of mean site data from Esh El Mellaha dikes

	Site	Sampling a/b	Declination	Inclination	α 95	K	R	Demagnetization
(a)	Acidic dikes							
	AD 4	no stable end point reached						
	AD 35	6/6	355.8	+59.2	6.8	97	5.948	25 mT, 450°C
	AD 42	6/6	5.2	+42.9	8.0	86	5.930	20 mT, 300°C
	AD 65A	6/6	356.9	+55.4	15.5	20	5.700	30 mT
	Mean		359.6	+49.3	10.9	(3 sites)		
	Paleopole		Lat. 87.3N	Long. 25.9E dm 14.5	dp 9.6			
(b)	Intermediate dikes							
	ID 26	5/7	10.1	+40.9	4.4	308	4.987	25 mT
	ID 27	5/6	12.7	+52.2	4.5	296	4.987	25 mT
	ID 28	5/6	13.9	+45.5	9.4	67	4.940	25 mT
	ID 29	no stable end point reached						
	ID 30	no stable end point reached						
	ID 31	8/9	177.4	−45.0	10.6	28	7.752	25 mT, 650°C
	ID 32	two stable components identified [see (d)]						
	ID 33	5/6	22.4	+47.8	13.4	33	4.880	10 mT, 400°C
	ID 34	3/3	2.1	+40.8	6.8	333	2.994	NRM
	ID 36	4/6	4.8	+46.6	8.6	115	3.974	15 mT, 250°C
	ID 37	6/6	6.0	+48.2	3.6	342	5.985	5 mT
	ID 41	8/10	5.3	+46.0	3.8	218	7.968	15 mT
	ID 46	two stable components identified [see (d)]						
	Mean	(excluding 31)	9.5	+46.2	4.1	(8 sites)		
	Paleopole		Lat. 81.6N	Long. 120.8E dm 5.2	dp 3.3			
(c)	Basic dikes							
	BD 1	6/8	344.2	+51.1	12.1	32	5.842	15 mT
	BD 2	7/7	7.5	+43.1	6.3	92	6.935	15 mT
	BD 3	6/6	348.2	+50.4	10.6	43	5.885	20 mT
	BD 38	7/8	0.5	+40.0	6.1	100	6.940	20 mT, 250°C
	BD 39	7/7	353.0	+46.7	4.8	162	6.963	15 mT, 250°C
	BD 40	7/7	5.5	+45.7	1.7	1200	6.995	25 mT, 400°C
	BD 43	7/7	353.4	+43.9	4.4	188	6.968	20 mT, 350°C
	BD 44	4/6	337.5	+41.3	7.7	143	3.979	25 mT
	BD 45	6/6	350.4	+40.9	9.5	61	5.902	NRM
	BD 69B	3/3	3.8	+12.0	4.6	735		20 mT, 450°C
	BD 74B	3/3	333.8	+25.3	55.9	6	2.7	30 mT, 500°C
	BD 6B	2/2	338.7	+37.3	—	—	—	25 mT
	BD 5	3/3	354.1	+28.0	32.3	16	2.6	20 mT
	Mean	(exluding 74B, 5)	353.3	+41.7	7.5	(11 sites)		
	Paleople		Lat. 82.7N	Long. 86.5W dm 9.1	dp 5.6			
(d)	Dikes with superposed components							
	Reversed magnetic component from sites 31, 32, 46	196.4		−46.3	24.7	26	2.923	3 sites (12 samples)
	Single samples from other sites normal component	353.0		+39.8	6.4	110	5.954	3 sites (6 samples)

Sampling: a, numbers used in calculations; b, number of samples examined; α 95 semi-angle of the cone of confidence at the 95% level and K a measure of dispersion (Fisher 1953); R, length of mean vector.

dikes and do not represent original magnetization. This provides one explanation of why the directions of magnetization of a wide variety of dikes of different ages is the same.

We have plotted the mean geomagnetic pole positions calculated for the dike sites on the apparent polar wandering curve of Vilas (1981) for western Gondwana. They fall at about the mid point of the 570–500 m.y. leg of the curve (Fig. 10). The antipoles also lie on the curve but near the end of the 1000–650 m.y. leg. We regard the former as the better interpretation of the age since the dikes penetrate both Younger Granites and rocks of the Dokhan volcanic formation, and thus the latter age implied is excessive.

The coincidence of the age implied by their location on the curve and our age estimates may be accidental, for calculation of the geomagnetic pole positions was made on the assumption that no tilt correction need be applied to the dikes. This may not be strictly warranted since the region lies close to the Gulf of Suez and the Esh El Mellaha Range may be a horst block or a titled fault block; as the dikes are vertical or nearly vertical, however, such a correction is probably not more than a few degrees. Also, the plane of the Nubian conformity is nearly horizontal and more recent movements have obviously had little effect.

CONCLUSIONS

When these data are taken in conjunction with the results from other dike swarms in parts of the Red Sea Hills far to the south they provide evidence for an extensive thermal event, a low-grade metamorphic event, occurring close to the end of the Cambrian, which reset the K/Ar ages and the remanent magnetization. It

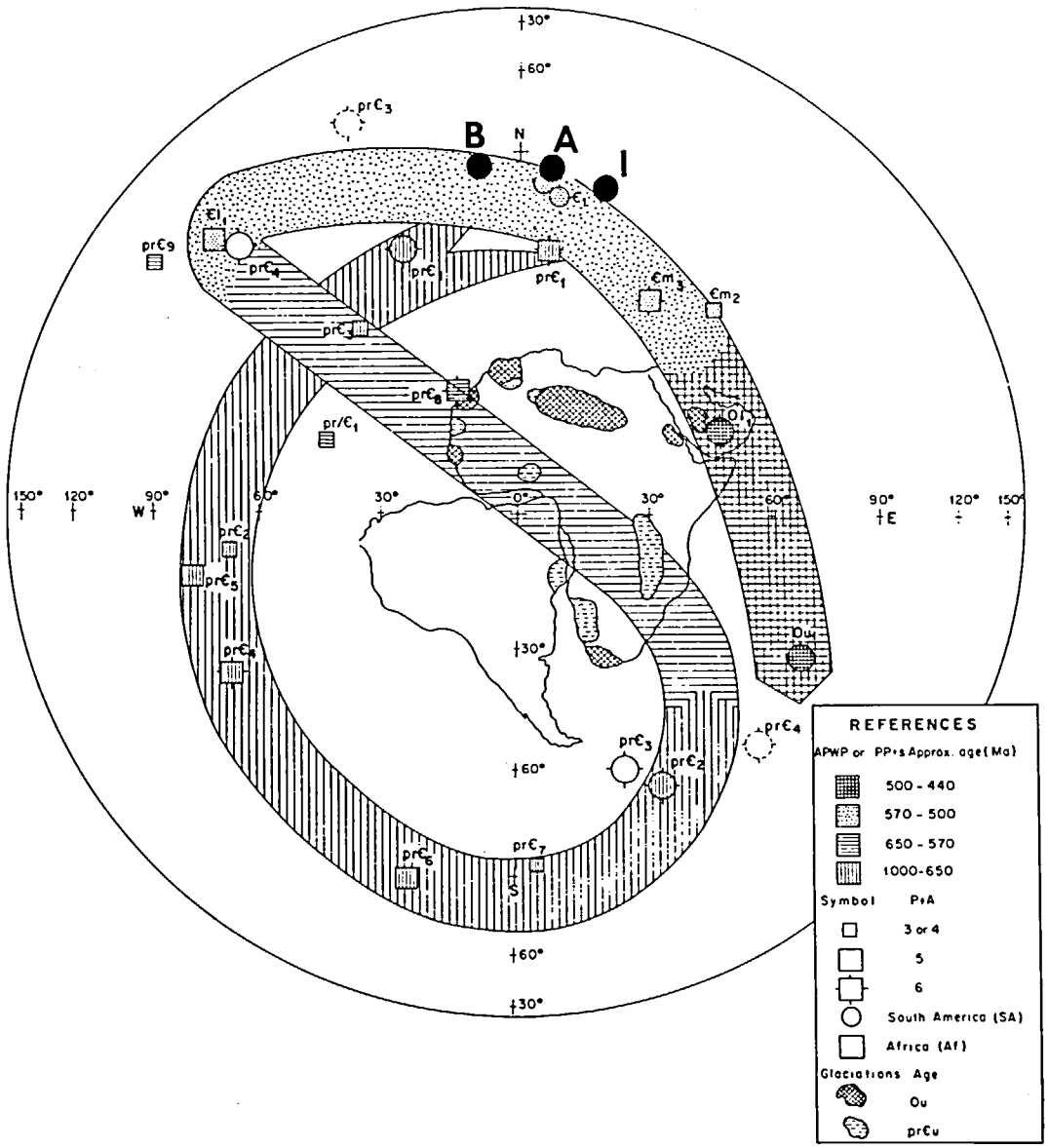


Fig. 10. Apparent polar wandering curve of Vilas (1981) on which are plotted geomagnetic pole positions for the Esh El Mellaha dikes: (A) acidic, (I) intermediate, (B) basic dikes.

suggests that, if less altered rocks can be found, it may be possible to retrieve a magnetic record of older events after removing the thermal imprint of the late Cambrian.

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